

**FIFTH SEMESTER U.G. DEGREE [SPECIAL] EXAMINATION
NOVEMBER 2020**

(CUCBCSS—UG)

Mathematics

MAT 5B 05—VECTOR CALCULUS

Time : Three Hours

Maximum : 120 Marks

Section A

Answer all questions.

Each question carries 1 mark.

1. Find the domain and range of $f(x, y) = \cos xy$.
2. Evaluate $\lim_{(x, y) \rightarrow (0, 0)} \frac{y}{x}$.
3. Define gradient of $\phi(x, y, z) = xyz$.
4. Compute the divergence of $\vec{f} = y^2 \vec{i} + z^2 \vec{j} + x^2 \vec{k}$.
5. What do you mean by the directional derivative of a function ?
6. When do you say that a vector field is conservative ?
7. What is the linearization of the function $f(x, y, z)$ at the point (x_0, y_0, z_0) ?
8. Find the total differential of u if $u = \ln(x^2 + y^2 + z^2)$.
9. What do you mean by circulation around a curve ?
10. State the Normal form of Green's theorem in the plane.
11. Fill in the blanks : If $\vec{a}$ is a constant vector and $\vec{r} = x \vec{i} + y \vec{j} + z \vec{k}$, then $\nabla \times (\vec{a} \times \vec{r}) = \text{_____}$.
12. State Stokes theorem.

(12 × 1 = 12 marks)

Turn over

Section B

Answer at least **eight** questions.

Each question carries 6 marks.

All questions can be attended.

Overall Ceiling 48.

13. Evaluate $\lim_{(x,y) \rightarrow (0,0)} \frac{x-y}{y+x}$.
14. Find the vector normal to the surface $\phi(x, y, z) = x^2 + y^2 + z^2$ at $(1, 1, 1)$.
15. Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ at $(1, 1, 1)$, if $x^2 + y^2 + z^2 + ye^x z + z \cos y = 0$.
16. Prove that $\nabla \cdot (r^n \vec{r}) = (n+3) \vec{r}$.
17. Find the total derivative of $u = x^3 + y^3$ with respect to t if $x = a \cos t$, $y = b \sin t$.
18. Compute the average value of the function $f(x, y) = x \cos(xy)$ over the rectangular region $0 \leq x \leq \pi$, $0 \leq y \leq 1$.
19. Find the directional derivative of $f(x, y) = xe^y + \cos(xy)$ at $(2, 0)$ in the direction of $3\vec{i} - 4\vec{j}$.
20. Find the flow of $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$ along the portion of the circular helix $x = \cos t$, $y = \sin t$, $z = t$; $0 \leq t \leq \pi/2$.
21. Find the value of λ which makes the following vector $\vec{f} = (\lambda xy - z^3)\vec{i} + ((\lambda - 2)x^2)\vec{j} + ((1 - \lambda)xz^2)\vec{k}$ is irrotational.
22. Verify whether the differential $e^x \cos y dx + (xz - e^x \sin y) dy + (xy + z) dz$ is exact or not.
23. If $\vec{f}$ and $\vec{g}$ are irrotational, then show that $\vec{f} \times \vec{g}$ is solenoidal.

24. Evaluate $\int_0^1 \int_0^2 xy(x-y) dy dx$.

25. Test whether the vector $\vec{f} = (e^x \cos y + yz)\vec{i} + (xz - e^x \sin y)\vec{j} + (xy + z)\vec{k}$ is conservative or not.

26. Prove or disprove : If $\text{div } \vec{f} = 0$, then $\text{curl } \vec{f} = 0$.

(8 × 6 = 48 marks)

Section C

Answer at least five questions.

Each question carries 9 marks.

All questions can be attended.

Overall Ceiling 45.

27. Evaluate $\int_0^a \int_y^a \frac{x}{x^2 + y^2} dy dx$.

28. Find $\nabla f(r)$, if $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$.

29. Find the work done by the force field $\vec{f} = yz\vec{i} + zx\vec{j} + xy\vec{k}$ along any of the path joining the points $(-1, 3, 9)$ and $(1, 6, -4)$.

30. Evaluate the outward flux of $\vec{f} = 4xz\vec{i} - y^2\vec{j} + yz\vec{k}$ through the surface of the cube cut from the first octant by the planes $x = 1, y = 1$ and $z = 1$.

31. Find the equation to the tangent plane and normal line to the surface $f(x, y, z) = x^2 + y^2 + z^2 - 9 = 0$ at the point $(1, 2, 4)$.

32. Evaluate the area enclosed by one leaf of the rose $r = 12 \cos 3\theta$.

33. Find the greatest and the smallest values of $f(x, y) = xy$ takes on the ellipse $x^2/8 + y^2/2 = 1$.

Turn over